PH.D. CANDIDATE · PURDUE UNIVERSITY

Magzhan.Mukanova

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Summary

I'm a Ph.D. candidate in the School of Applied and Creative Computing at Purdue University, researching AI-driven embodied virtual agents at the intersection of computer graphics, animation, and human-computer interaction. My focus is on familiarity and familiar virtual agents: how an agent's appearance, voice, and motion can be designed to feel recognizable or identity-linked, and how that familiarity shapes trust, comfort, and perception. This work draws on behavioral animation, 3D character modeling, and AI systems that drive an agent's gesture, posture, and facial behavior, evaluated through both animation practice and empirical HCI methods.

Education

Ph.D. in Computer and Information Technology

EXPECTED MAY 2027

Purdue University — School of Applied and Creative Computing · West Lafayette, IN

FOCUS Research on embodied virtual agents, digital clones, conversational AI, and animation perception.

M.S. in Computer Graphics Technology

2023

Purdue University · West Lafayette, IN

GPA 3.98 · THESIS *Affective Arm and Hand Gestures in Animated Pedagogical Agents*

Research Projects

Perceptions of Familiarity in Virtual Agents

ONGOING

Ph.D. research

- Study how familiarity in a virtual agent's appearance, voice, and motion shapes how people perceive it (ACM SAP 2026).
- Build digital-clone and MetaHuman agents and design user perception studies around them.

Conversational Embodied Agent and VR

ONGOING

Unreal Engine 5/Unity · Conversational AI

- Built an interactive embodied agent that holds real-time conversations, combining speech, gaze, facial animation, and gesture behavior.

Affective Gestures for Pedagogical Agents

2023

M.S. thesis · ArtsIT 2023

- Animated pedagogical agents performing affective arm and hand gestures extracted from the GEMEP dataset.
- Ran a perception study with 131 participants: recognition of specific emotions was low, but viewers identified arousal and valence more accurately.

Teaching & Experience

Teaching Assistant

AUG 2024 – PRESENT

Purdue University · West Lafayette, IN

- Teach CGT 123: Introduction to Animation, covering 2D/3D animation principles, timing, spacing, acting, and character performance.
- Support students with Maya, Unreal Engine, iClone Reallusion, and motion-capture and AI workflows.
- Help students troubleshoot rigging, keyframing, mocap cleanup, and rendering in hands-on labs.
- Demonstrate facial animation, body mechanics, gesture work, and scene staging.
- Teaching assistant to CGT 342 Advanced 3D animation, CGT 442 Production for Computer Animation, CGT44201 Advanced Portfolio Development for Digital Media Careers.

3D Animator

APR 2024 – NOV 2025

Everstar Studio · Remote, Canada

- Created facial and body animation for virtual characters, including motion capture cleanup and polishing.

- Worked with MetaHumans, iClone, and procedural animation systems in Unreal Engine.
- Integrated AI-assisted tools into the animation pipeline to improve realism and production efficiency.

Mentor
DEC 2023 – APR 2026

BrainCode Academy · Remote, Astana, KZ

- Advised game dev course students on UE5, environment design, Blender, and 3D modelling; invited industry professionals to online meetings with students.

Editor
JUN 2017 – APR 2019

Khabar TV agency· Astana, KZ

- Organized, reviewed, cataloged, and maintained audiovisual archives, including news footage and broadcast recordings, ensuring accurate records, efficient retrieval, and preservation of media assets.

Assistant Director of Broadcasting
JUN 2015 – JUN 2017

Khabar TV agency· Astana, KZ

- Coordinated broadcasting operations, production schedules, and program preparation while supporting directors, managing production workflows, and collaborating with staff to ensure smooth and timely broadcast delivery.

Publications

Perceptions of Appearance, Voice, and Motion Familiarity in Virtual Agents
2026

Mukanova, M., Yang, F.-C., Adamo, N., & Mousas, C. · *ACM Symposium on Applied Perception 2026 Article No.: 1, Pages 1 - 17*

Recognizing and Relating to the Race/Ethnicity and Gender of Animated Pedagogical Agents
2024

Zhao, F., Mayer, R. E., Adamo-Villani, N., Mousas, C., Choi, M., Lam, L., **Mukanova, M.**, & Hauser, K. · *Journal of Educational Computing Research, 62(3), 455–481*

Animated Pedagogical Agents Performing Affective Gestures Extracted from the GEMEP Dataset: Can People Recognize Their Emotions?
2024

Mukanova, M., Adamo-Villani, N., Mousas, C., Zhao, F., Choi, M., Lam, L., & Hauser, K. · *ArtsIT 2023, Springer LNICST vol. 565*

Effects of Body Type and Voice Pitch on Perceived Audio-Visual Correspondence and Believability of Virtual Characters
2023

Lam, L., Choi, M., **Mukanova, M.**, Hauser, K., Zhao, F., Mayer, R., Mousas, C., & Adamo-Villani, N. · *ACM Symposium on Applied Perception 2023, Article 3*

Affective Arm and Hand Gestures in Animated Pedagogical Agents
2023

Mukanova, M. · *M.S. Thesis, Purdue University (ProQuest)*

Service & Volunteering

Reviewer — IEEE VR
IEEE Conference on Virtual Reality and 3D User Interfaces

Counselor-Researcher — Counterventions Summer Camp
JUL 2026

CoLiberation Lab, Purdue University

- Five-day robotics summer camp for autistic children.

UN Volunteer — Animation for Social Change
JUN 2023

Simuke Africa Youth Association, Zimbabwe

- Created an animation video supporting the campaign to end child marriage in Zimbabwe.

ALSO Managed UNICEF Kazakhstan “Plastic Free” project; 5 years of volunteering at Best for Kids Public Fund.

Skills & Languages

TECHNICAL VR, Unity, Vicon Motion Capture, Markerless Motion Capture, Unreal Engine 5, MetaHumans, Autodesk Maya, Blender, iClone, Character Creator, motion capture pipelines, procedural animation, AI mocap tools, Adobe Creative Suite, Python, OpenGL

RESEARCH User perception studies, experimental design, data analysis in Python

LANGUAGES Kazakh (native), Russian (native), English (advanced), German (beginner)